Linear Graphs 5: Parallel Lines

Name _____ **Date** _____ No calculator Version 1

KEY WORDS gradient · y-intercept · y = mx + c · two points · parallel · x-intercept · perpendicular

A Gradient from two points ●○○

WORK IN YOUR BOOK

Remember: Gradient $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Keep the points in the same order on the top and the bottom.

1. Find the gradient of the line through each pair of points.

a) $(1,\ 3)$ and $(4,\ 9)$

b) $(0,\ 5)$ and $(2,\ 1)$

c) $(-2,\ 1)$ and $(4,\ 4)$

d) $(3,\ -2)$ and $(5,\ 8)$

e) $(-1,\ 6)$ and $(3,\ -2)$

f) $(-4,\ -3)$ and $(2,\ 0)$

B From a gradient and a point ●●○

WORK IN YOUR BOOK

Remember: Put the gradient and the point into $y = mx + c$ and solve for c. Then write the equation.

2. Find the equation of the line with the given gradient that passes through the given point.

a) Gradient 2, through $(1,\ 5)$

b) Gradient 3, through $(2,\ 1)$

c) Gradient -1, through $(4,\ 2)$

d) Gradient -2, through $(-1,\ 6)$

e) Gradient $\dfrac{1}{2}$, through $(4,\ 5)$

f) Gradient $-\dfrac{2}{3}$, through $(3,\ -1)$

C From two points ●●○

WORK IN YOUR BOOK

Remember: Find the gradient first. Then use either point to find c. Check your equation with the other point.

3. Find the equation of the line through each pair of points.

a) $(1,\ 3)$ and $(3,\ 7)$

b) $(0,\ 5)$ and $(2,\ 1)$

c) $(2,\ 1)$ and $(4,\ 2)$

d) $(-1,\ -5)$ and $(2,\ 4)$

e) $(-2,\ 6)$ and $(2,\ -2)$

f) $(-3,\ -1)$ and $(3,\ 3)$

D Parallel lines ●●●

WORK IN YOUR BOOK

Remember: Parallel lines have the same gradient. If an equation is not in the form $y = mx + c$, rearrange it before you read off the gradient.

4. Find the equation of each line.

a) Parallel to $y = 3x - 1$, through $(2,\ 10)$

b) Parallel to $y = -2x + 5$, through $(-1,\ 4)$

c) Parallel to $2y = 6x + 2$, through $(1,\ 2)$

d) Parallel to $x + 2y = 6$, through $(4,\ -1)$

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5. Are these pairs of lines parallel? Show how you know.

a) $y = 2x + 3$ and $2y = 4x - 2$

b) $3x + y = 7$ and $y = 3x - 2$

c) $x - 2y = 4$ and $2y = x + 6$

d) $y = 4 - x$ and $x + y = 1$

E Checking points and finding intercepts ●●● WORK IN YOUR BOOK

> **Remember:** To test a point, put its x into the equation and see whether you get its y. A line crosses the x-axis where $y = 0$.

6. Is each point on the line? Show your working.

a) Is $(4,\ 11)$ on the line $y = 2x + 3$?

b) Is $(2,\ 1)$ on the line $y = -3x + 5$?

c) Is $(6,\ -1)$ on the line $y = \frac{1}{2}x - 4$?

7. Find the equation of the line through each pair of points, then the point where it crosses the x-axis.

a) $(1,\ 1)$ and $(3,\ 5)$

b) $(-2,\ 8)$ and $(2,\ 0)$

c) $(0,\ 3)$ and $(6,\ 6)$

★ Extension FOR EARLY FINISHERS

8. Two lines are perpendicular (at right angles) when their gradients multiply to -1. A line has gradient 2. What is the gradient of a line perpendicular to it?

9. Find the equation of the line through $(2,\ 1)$ that is perpendicular to $y = \frac{1}{2}x + 3$.

10. A is $(1,\ 2)$ and B is $(5,\ 4)$.

a) Find the midpoint of AB.

b) Find the equation of the perpendicular bisector of AB: the line through the midpoint at right angles to AB.

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Linear Graphs 5: Parallel Lines

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS gradient · y-intercept · y = mx + c · two points · parallel · x-intercept · perpendicular

A Gradient from two points ●○○

WORK IN YOUR BOOK

Remember: Gradient $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. Keep the points in the same order on the top and the bottom.

1. Find the gradient of the line through each pair of points.

a) $(1,\ 2)$ and $(3,\ 10)$

b) $(0,\ 7)$ and $(3,\ 1)$

c) $(-3,\ 2)$ and $(3,\ 5)$

d) $(2,\ -5)$ and $(4,\ 7)$

e) $(-2,\ 5)$ and $(2,\ -3)$

f) $(-6,\ -1)$ and $(2,\ 3)$

B From a gradient and a point ●●○

WORK IN YOUR BOOK

Remember: Put the gradient and the point into $y = mx + c$ and solve for c. Then write the equation.

2. Find the equation of the line with the given gradient that passes through the given point.

a) Gradient 3, through $(1,\ 7)$

b) Gradient 2, through $(3,\ 1)$

c) Gradient -1, through $(5,\ 1)$

d) Gradient -3, through $(-1,\ 4)$

e) Gradient $\dfrac{1}{2}$, through $(6,\ 7)$

f) Gradient $-\dfrac{2}{3}$, through $(6,\ -2)$

C From two points ●●○

WORK IN YOUR BOOK

Remember: Find the gradient first. Then use either point to find c. Check your equation with the other point.

3. Find the equation of the line through each pair of points.

a) $(1,\ 5)$ and $(3,\ 11)$

b) $(0,\ 4)$ and $(3,\ -2)$

c) $(2,\ 3)$ and $(6,\ 5)$

d) $(-1,\ -7)$ and $(2,\ 5)$

e) $(-3,\ 5)$ and $(1,\ 1)$

f) $(-3,\ 1)$ and $(3,\ 5)$

D Parallel lines ●●●

WORK IN YOUR BOOK

Remember: Parallel lines have the same gradient. If an equation is not in the form $y = mx + c$, rearrange it before you read off the gradient.

4. Find the equation of each line.

a) Parallel to $y = 2x + 7$, through $(3,\ 1)$

b) Parallel to $y = -3x - 2$, through $(-1,\ 5)$

c) Parallel to $2y = 4x - 6$, through $(2,\ 1)$

d) Parallel to $x + 3y = 9$, through $(6,\ -2)$

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5. Are these pairs of lines parallel? Show how you know.

a) $y = 3x - 2$ and $3y = 9x + 6$

b) $2x + y = 5$ and $y = 2x + 1$

c) $x - 3y = 6$ and $3y = x + 9$

d) $y = 5 - x$ and $x + y = 2$

E Checking points and finding intercepts ●●● WORK IN YOUR BOOK

> **Remember:** To test a point, put its x into the equation and see whether you get its y. A line crosses the x-axis where $y = 0$.

6. Is each point on the line? Show your working.

a) Is $(4,\ 10)$ on the line $y = 3x - 2$?

b) Is $(3,\ -1)$ on the line $y = -2x + 7$?

c) Is $(8,\ 7)$ on the line $y = \frac{1}{2}x + 3$?

7. Find the equation of the line through each pair of points, then the point where it crosses the x-axis.

a) $(1,\ 3)$ and $(3,\ 7)$

b) $(-1,\ 7)$ and $(3,\ -1)$

c) $(0,\ 2)$ and $(4,\ 4)$

★ Extension FOR EARLY FINISHERS

8. Two lines are perpendicular (at right angles) when their gradients multiply to -1. A line has gradient 3. What is the gradient of a line perpendicular to it?

9. Find the equation of the line through $(3,\ 1)$ that is perpendicular to $y = \frac{1}{3}x + 2$.

10. A is $(2,\ 1)$ and B is $(6,\ 3)$.

a) Find the midpoint of AB.

b) Find the equation of the perpendicular bisector of AB: the line through the midpoint at right angles to AB.

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